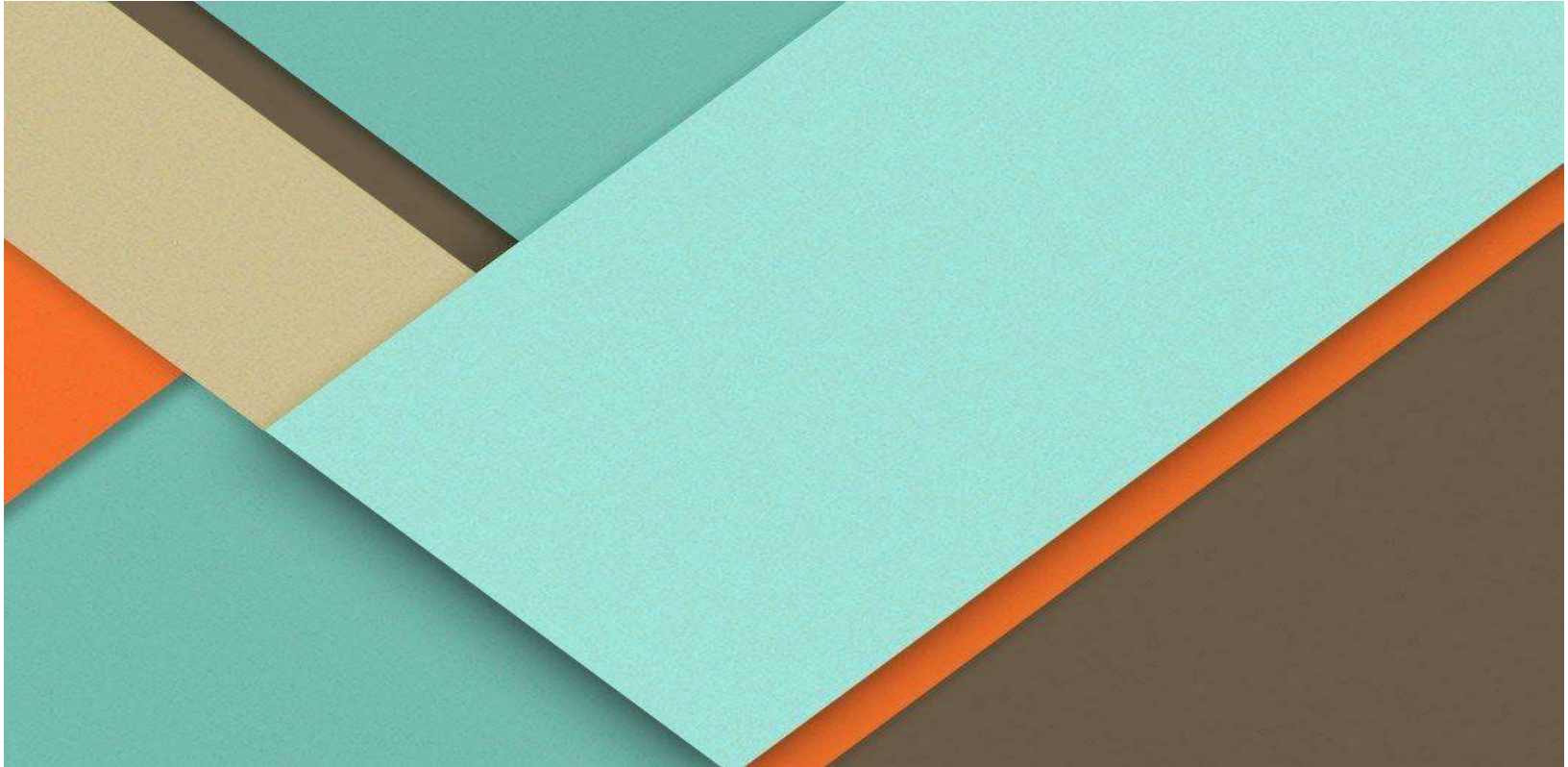




ENGINEERING PORTFOLIO

Karnan Thamilchelvan

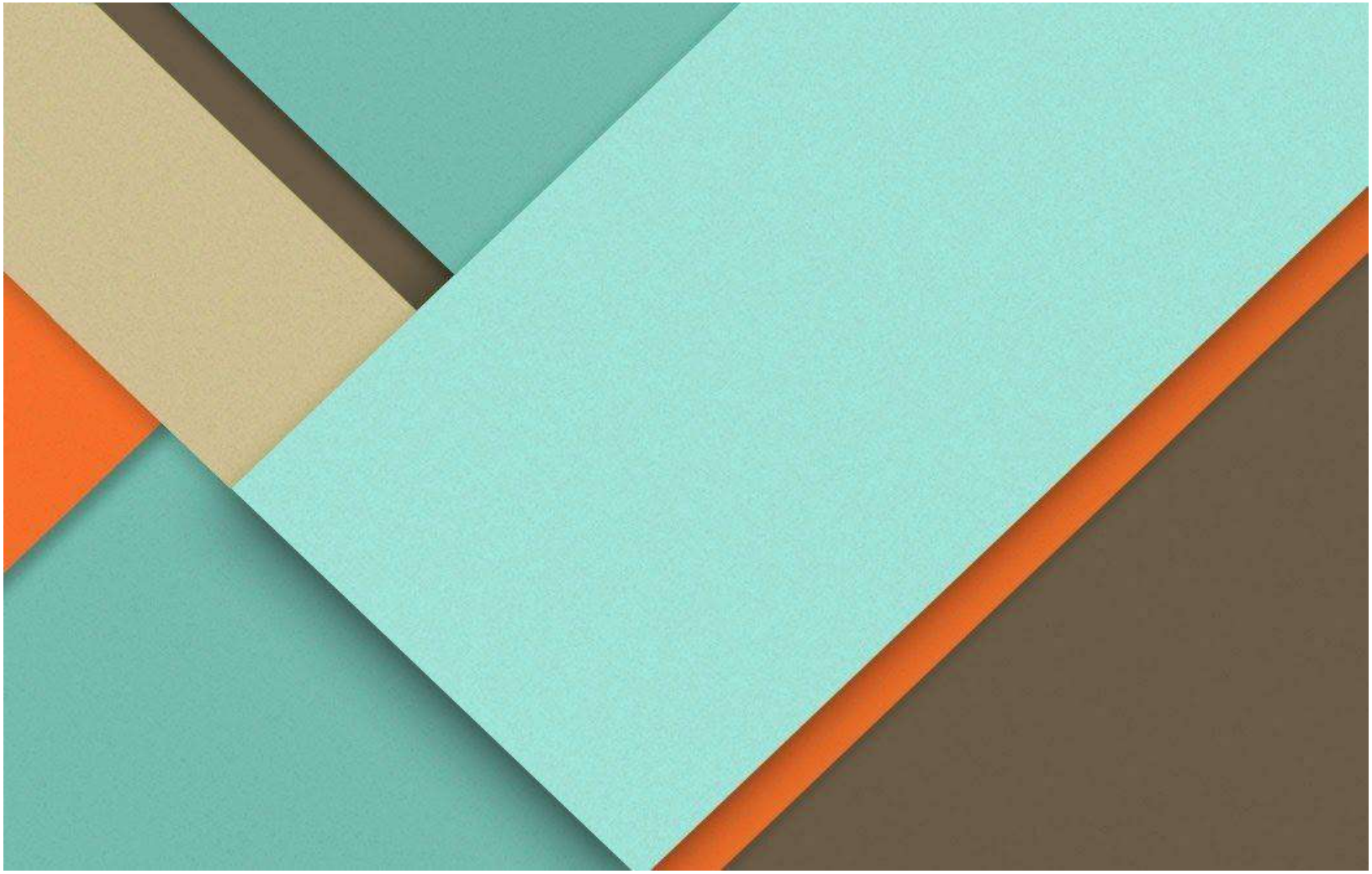


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Profile



About this portfolio

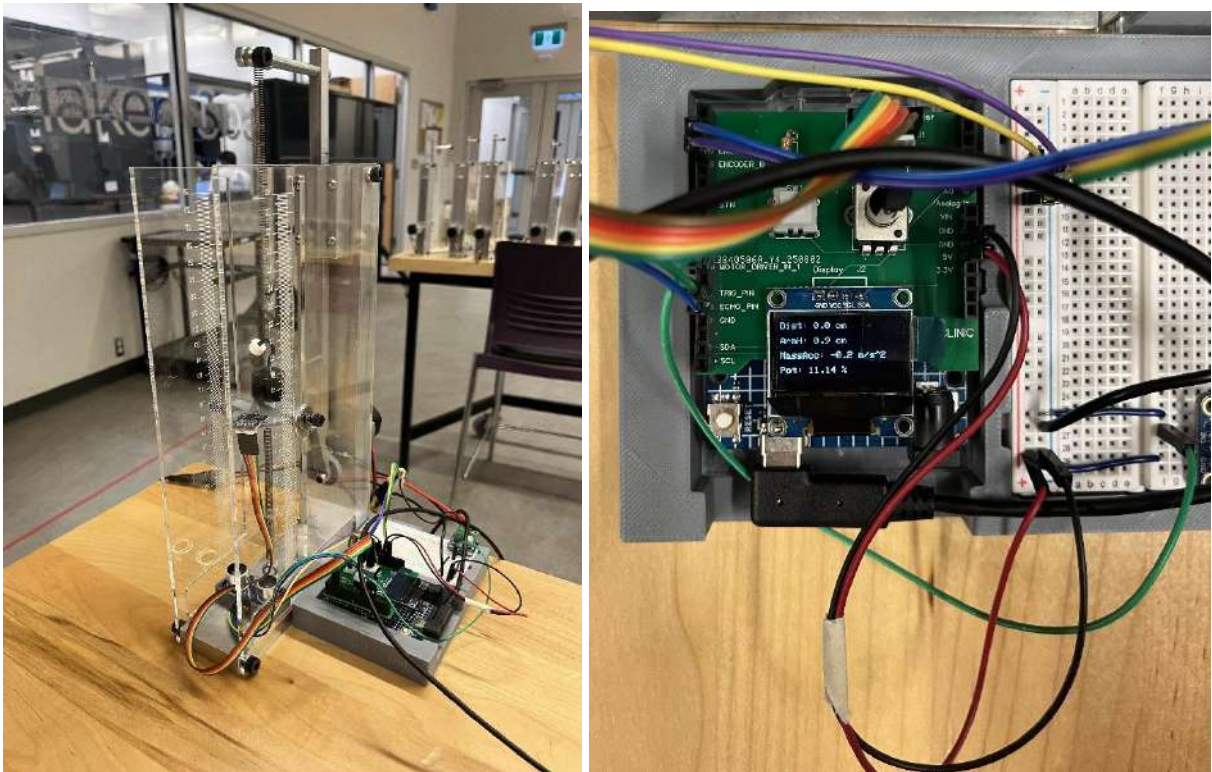
I'm an undergraduate Computer Engineering student passionate about hands-on hardware, embedded systems, and software projects. This portfolio highlights select work demonstrating practical problem-solving, design, and technical implementation. For questions or collaboration, feel free to reach out via email.

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Mechatronic Resonance System

Overview ([Mechatronic Resonance Github](#))

The Mechatronic Resonance System is a mass-spring-damper platform designed for the University of Waterloo's ECE198 Dynamic Systems Lab. It combines mechanical design, custom electronics, and embedded programming to help students explore resonance, oscillations, and applied circuitry.



Project Highlights

- **Arduino UNO R4 WiFi** microcontroller for real-time motor control and sensor interfacing.
- **Custom PCB** designed in EasyEDA, fabricated, and hand-soldered to streamline sensor connections and ensure reliable integration.
- **Motion and proximity sensors** capture oscillation data, enabling live resonance demonstrations.
- Modeled in SolidWorks and manufactured **over 220+ 3D-printed and machined parts**, including casings, bearings, and mounts.
- Fabricated and assembled **40 complete units**.



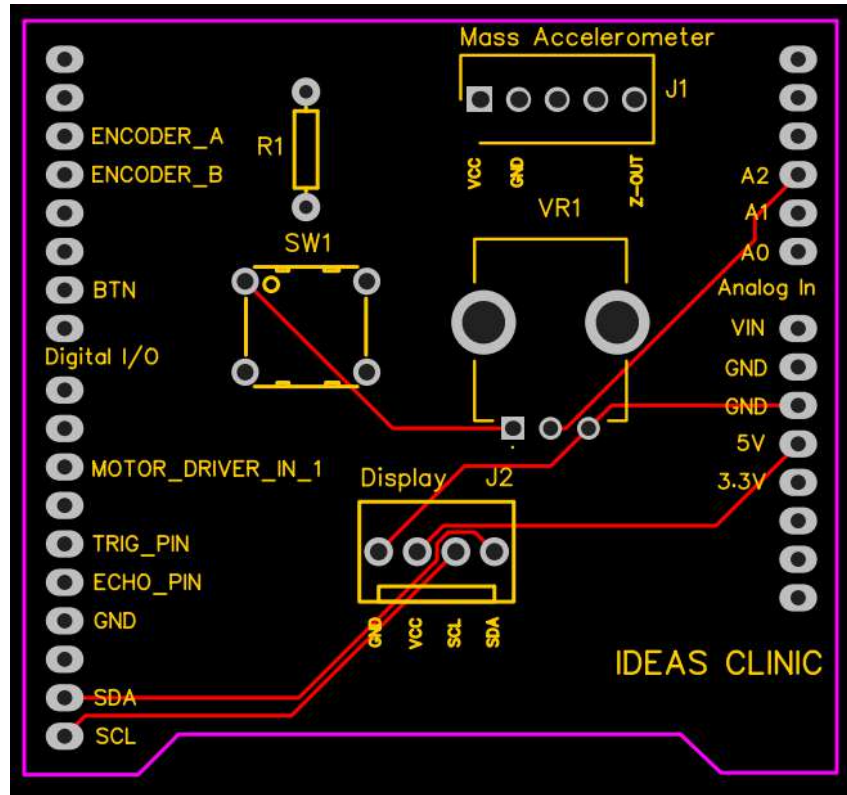
System Operation

- The motor excites the spring-mass system at varying frequencies, allowing students to observe resonance and damping behavior.
- Motion and proximity sensors measure oscillations in real time, sending data to the microcontroller.

PCB Design and Integration

- Custom PCB layout created in EasyEDA to reduce wiring complexity and improve durability for repeated lab use.
- Components such as headers, potentiometers, and displays were hand-soldered, ensuring robust connections for sensors and motor drivers.

- The PCB served as a compact interface between the Arduino UNO R4, sensors, and motors.



Smart-Irrigation Controller

Overview ([Smart-Irrigation Github](#))

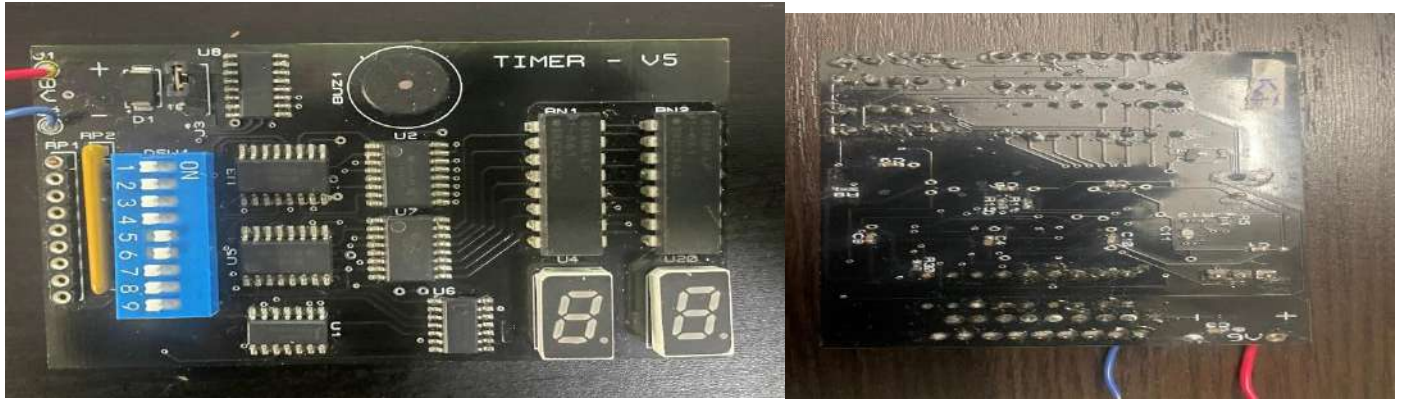
The Smart-Irrigation Controller is an embedded system that enhances water and energy efficiency in terraced vineyards. By addressing real-world agricultural challenges, this project integrates technology and sustainability to optimize irrigation practices.

Project Highlights

- Servo Mechanism:** Directs water flow to specific zones, indicated by RGB LED signals.
- PWM-Controlled Pump:** Adjusts motor speed for optimal water pressure across zones. Ensures precise water distribution to irrigation zones.
- Ultrasonic Sensor:** Monitors reservoir levels by providing accurate water depth measurements. These measurements are sent via UART communication.
- Seven-Segment Display:** Displays real-time water level percentages for easy monitoring.
- Potentiometer:** Enables manual adjustment of pump RPM as needed.
- MCU Timer:** Simulates real-time operations on a scaled timeline.
- STM32 Nucleo Development Board:** Serves as the central platform for connecting and integrating all sensors and actuators in the project.

MCU Timer Board

- Assembly and testing of the timer board were critical aspects of this project, ensuring accurate and reliable system functionality.
- Components included Op-amps, ICs, digital displays, resistors, switches, diodes, etc.
- Components were soldered onto the circuit board for durability.



System Operation

Filling the Reservoir:

- Overnight refill is managed by an inlet pipe and servo, with status indicated by a RGB LED.

Irrigation Process:

- Distributes water to three irrigation zones based on a programmed schedule.
- Pump speed is dynamically adjusted via PWM to meet the specific needs of each zone:
- **Zone 1:** 50,000 gallons/day, 25-foot HEAD.
- **Zone 2:** 30,000 gallons/day, 50-foot HEAD.
- **Zone 3:** 14,000 gallons/day, 60-foot HEAD.
- **Inlet:** 40-foot HEAD from the spring.

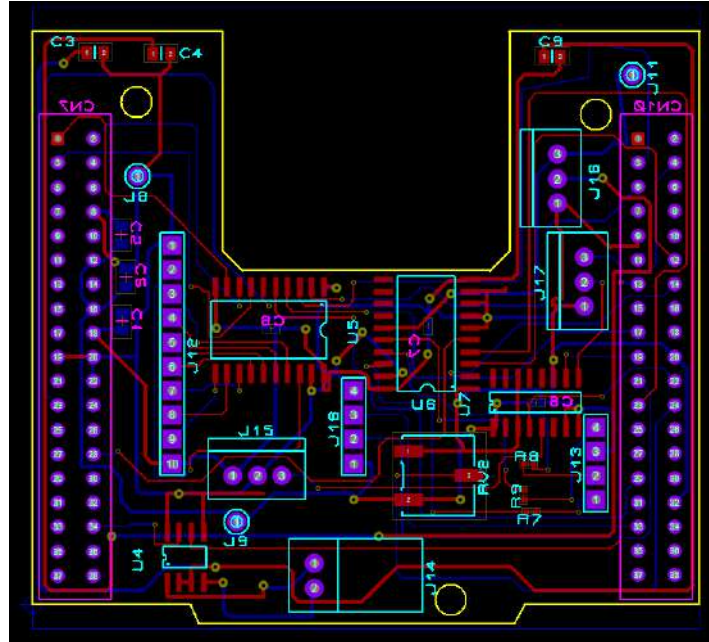
Overall System

- The Smart-Irrigation Controller combines sensors, actuators, and a microcontroller to automate irrigation. The timer board schedules operations, while the microcontroller processes inputs and controls components like the pump and display. Together, these elements ensure efficient water management and energy optimization.



PCB Design and Integration

- To finalize the Smart-Irrigation Controller, a custom PCB was designed using the Proteus PCB Design Tool. This PCB integrated the STM32 microcontroller and the Timer Board into a compact and reliable layout, reducing wiring complexity and enhancing system efficiency.
- Below is an image of the PCB design:



RainSense: Smart Water Management System

Overview ([RainSense GitHub](#))

RainSense is a smart water management solution designed to help farmers optimize irrigation and reduce water bills. By combining IoT technology, environmental sensors, and weather data, the system provides real-time insights and actionable notifications on crop irrigation, water usage, and rainwater harvesting.

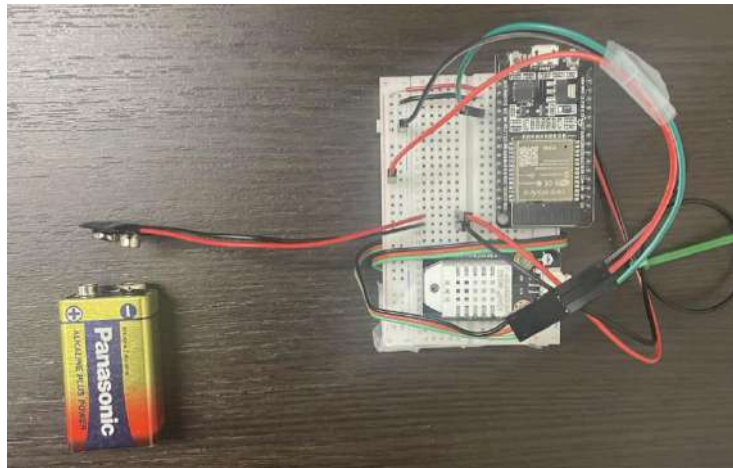
Project Highlights

- **Weather-Based Irrigation Notifications:**
 - Fetches weather data via API to determine optimal watering times. Uses Wi-Fi protocol to send API weather data.
 - Alerts users of incoming rainstorms to prepare rainwater collectors.
- **Environmental Monitoring:**
 - Measures real-time local conditions using temperature, humidity, and pressure sensors.
 - Cross-references local sensor data with weather forecasts for accurate recommendations.
- **Location-Based Insights:**

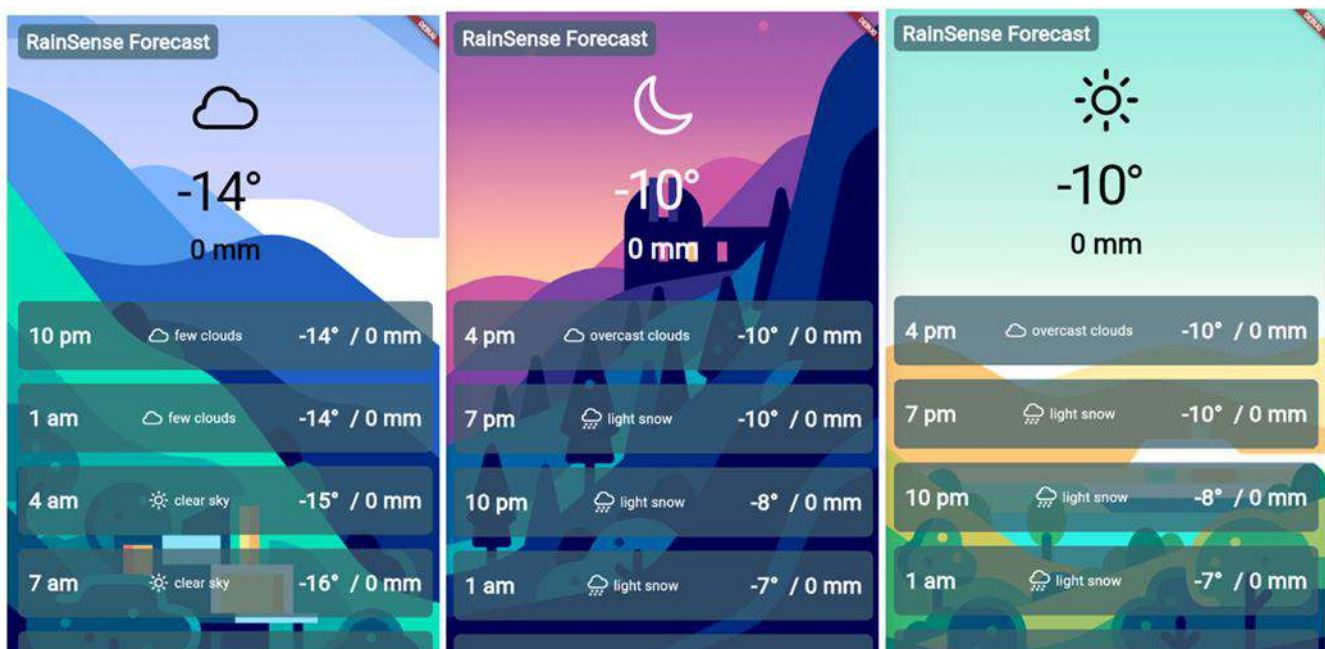
- Customizable location input for region-specific weather updates and irrigation suggestions.
- **Water Tank Management:**
 - Monitors water tank levels with a distance sensor.
 - Displays real-time tank capacity and alerts for low or full levels.
 - Estimates water usage and potential cost savings.

System Components

- **Hardware:**
 - Sensors: Temperature, humidity, pressure, and distance (for tank monitoring)
 - Microcontroller: ESP32 for data processing and Wi-Fi protocol communication.
 - Rainwater harvesting system.



- **Software:**
 - Integration with a weather API for real-time and forecasted data.
 - Mobile app for monitoring, notifications, and water usage tracking.



How it Works

1. Weather Data Integration:

- The system retrieves forecasted and real-time weather data based on user location.

2. Sensor Monitoring:

- Environmental sensors measure local temperature, humidity, and pressure.
- A distance sensor monitors water tank levels.

3. Decision Making:

- Combines forecasted and real-time data to:
- Notify users of optimal irrigation schedules.
- Alert users about upcoming rainstorms.

4. Water Tank Monitoring:

- Measures tank levels and warn users when levels are critically low or full.
- Tracks water usage and calculate cost savings.

5. User Notifications:

- Sends irrigation schedules, tank status updates, and rain alerts via the mobile app.

Project Goals

- Optimize irrigation to reduce water bills.
- Encourage sustainable rainwater harvesting practices.
- Provide farmers with real-time insights and actionable data for efficient water management.

